Market Efficiency, Short Sales Costs & Constraints, and Trading Volume; A Structural Study

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March 26, 2020

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[2] I am grateful to Reena Aggarwal, Matt Armstrong, Ray Ball, Sudipta Basu, Martijn Cremers, Kamalini Ganguly, David Hirshleifer, Michael McDonald, Stefan Nagel, Raghu Sundaram, Erik Sirri, David Weinbaum, Simon Wheatley (in alphabetical order), and especially Jeffrey Pontiff and Hersh Shefrin, for their very helpful thoughts and suggestions. The usual caveats apply.

Abstract

I use eight metrics as separate negative measures for market efficiency. I develop a theory of trading volume as a function of short sales costs and market efficiency as a function of trading volume. I identify instruments for the endogenous variables and for those measured with error. I use 3SLS and EIV to estimate this two-equation structural model and test the one-tailed hypotheses, separately for Nasdaq and non-Nasdaq U.S. stocks. Accounting for Fama-French factors and institutional ownership, I find that the impact on market efficiency of short sales costs & constraints is *not* negative and that of trading volume is *not* positive.

Keywords: Event Study; Earnings Announcements; Key Developments; Arbitrage Risk; Idiosyncratic Risk; Endogeneity; Simultaneity; Proxy Variables; Instrumental Variables.

JEL Classification Codes: G14; G12; C58; C33; C36.

1 Introduction

I use eight different metrics as separate measures of the efficiency of the market for a stock and describe how they all relate to one another. I develop a simple two-equation structural model that shows that the impacts on market efficiency of a) short sales costs & constraints and b) trading volume are theoretically indeterminate and are, therefore, empirical questions. I use the panel nature of the data to identify appropriate instruments for the endogenous variables and for the variables that are measured with error, and I use Three-Stage Least Squares and Errors in Variables to estimate this structural model and test the one-tailed hypotheses, accounting for the Fama-French Factors (market cap, leverage, book-to-market equity ratio, and price-to-earnings ratio) and institutional ownership. Analyzing Nasdaq stocks and non-Nasdaq stocks separately,[1] I find that, contrary to *much* previous theoretical and empirical work, *ceteris paribus*, the efficiency of the market for the stock is *not* negatively impacted by short sales costs & constraints and is *not* positively impacted by normalized trading volume.

The economic theory and empirics of this paper can be summarized as: Separately for Nasdaq and non-Nasdaq stocks, and with and without controls for Fama-French Factors and institutional ownership, normalized trading volume is negatively and significantly impacted by short sales costs & constraints, and trading volume does *not* positively and significantly impact market efficiency, and therefore, the impact on market efficiency of short sales costs & constraints is *not* negatively significant.

A market is *semistrong efficient* if prices reflect all publicly available information — see, for example, Fama (1970), Malkiel (1992), Harris (2003), Shleifer

[1] Using a number of different variants of the metrics, different specifications of the regression equations, and different instrumentation strategies, and using data for **all** the full years for which I have complete data.

(2000), and Samuelson (2009). As has been recognized, by, for example, Fama, Fisher, Jensen and Roll (1969), a market is efficient if "stock prices adjust very rapidly to new information." Finally, as noted by, for example, Fama (1970), the notion of market efficiency is that the "total excess market value at $t+1$ that will be generated by such a system ... which, from the 'fair game' property ... has expectation ... 0," in other words, that abnormal returns, on average, are zero in an efficient market. 1) Prices of securities reflect, *albeit to varying extents*, all publicly available information, 2) prices adjust, *albeit to varying extents*, to new information, and 3) abnormal returns are close to zero, also *albeit to various extents* — therefore, markets for securities are efficient in varying degrees — often referred to as *relative efficiency,* see, for example, Campbell, Lo and MacKinlay (1997).[2] In this paper, I use eight metrics as *separate and objective negative measures* of the efficiency of the market for a stock — see Section 6.[3,4]

Lamont and Thaler (2003) say that "short sale constraints have long been recognized as crucial to the workings of efficient markets." For the importance of short sales costs & constraints, see most importantly, Miller (1977), and, more recently (in reverse chronological order), Drechsler and Drechsler (2016), Muravyev, Pearson, and Pollet (2016), Boehmer and Wu (2013), Saffi and Sigurdsson (2011), Bris, Goetzmann and Zhu (2007), Bai, Chang and Wang (2006), Boehme, Danielsen and Sorescu (2006), Lamont and Stein (2004), Duffie, Garleanu and Pedersen (2002), Jones and Lamont (2002), Dechow, Hutton, Meulbroek and Sloan (2001), Easley, O'Hara and Srinivas (1998), Duffie (1996), and others. However, Diamond and Verrecchia (1987) study a model where short sales constraints do not affect market efficiency. Rapach, Ringgenberg and Zhou

[2] For a description of the importance of relative efficiency for valuations (especially Mark-to-Market) and securities class actions (especially class certification), see Bhattacharya (2019).

[3] See Appendix 1 for summary statistics, especially correlation coefficients, of all these eight different measures of market efficiency that I use in this paper.

[4] See Section 9 on other measures of market efficiency that I intend to study in the future.

(2016) “show that short interest, aggregated across securities, is arguably the strongest predictor of the equity risk premium identified to date.” Boehmer, Huszar and Jordan (2010) show that “the positive information associated with low short interest, which is publicly available, is only slowly incorporated into prices, which raises a broader market efficiency issue. Our results also cast doubt on existing theories of the impact of short sale constraints.” In this paper, I use normalized short interest as a negative proxy for short sales costs & constraints — this is based on the *Law of Demand* hypothesis that, *ceteris paribus*, the short interest for a stock (the quantity demanded) is a decreasing function of the short sales costs and constraints (price), since, as Lamont and Thaler (2003) show, the supply is fairly constant as a function of price.

The relation of trading volume with market efficiency has been studied by, for instance, Bhattacharya and O'Brien (2015), O'Hara (1997), Stickel and Verrecchia (1994), Blume, Easley and O'Hara (1994), Wang (1994), and Karpoff (1987).[5] In particular, Bhattacharya and O'Brien (2015) estimated a reduced form, and using arbitrage risk as the negative measure of market efficiency, came to the conclusion that market efficiency was not significantly positively related to trading volume — see also the extensive discussion there on the “conventional wisdom” regarding the dependence of market efficiency on trading volume.

In this paper, I develop a simple two-equation structural system. In the first equation, I model trading volume (and number of market makers, for Nasdaq stocks) as functions of dispersion of investor valuations, investor holding probability, transaction costs, and short sales costs. In the second equation, I model, on the basis of information economics, market efficiency as a function of normalized trading volume and number of security analysts following the stock.[6] I

[5] Beaver (1968) refers to the “economist's notion that volume reflects a lack of consensus regarding the price.”

[6] The Fama-French Factors (market cap, leverage, book-to-market equity ratio, and price-to-earnings ratio) and institutional ownership are not directly in the theoretical model because there are no convincing reasons, *a priori*. However, as I explain later, I do account for these

show that higher trading volume will be observed with lower short sales costs, lower transaction costs, higher dispersion of investor valuations, and/or higher likelihood that an investor holds the asset. However, for instance, there is no reason that higher dispersion in investor valuations would lead to higher market efficiency — I show an example in Subsection 4.1.1 where there is no dispersion in valuations and the market is perfectly semistrong form efficient, but with no trades. A corollary of my two-equation system is that short sales costs & constraints impact trading volume, that in turn impacts market efficiency. Given the above discussion, the impact on market efficiency of normalized trading volume, everything else remaining the same, is fundamentally an empirical question, and therefore, the impact on market efficiency of short sales costs, everything else remaining the same, is also an empirical question.

Appendix 1 provides summary statistics of the different negative measures of market efficiency and how they relate to one another. Section 2 details event studies. Section 3 discusses arbitrage risk and idiosyncratic risk. Section 4 develops a simple structural model of trading volume and market efficiency. Section 5 describes the data and Appendix 2 summarizes the relationship between the potential factors of market efficiency studied in this paper. Section 6 lists the measures of market efficiency used in this paper. Section 7 describes the econometric methodology of this paper. Section 8 presents the meta-summaries of the thousands of regressions that were part of the main analyses and the additional tens of thousands of regressions that were run to check for robustness. Section 9 concludes and provides directions for future research.

factors explicitly in my empirical analyses.

2 Event Studies

Since Fama, Fisher, Jensen and Roll (1969) and Ball and Brown (1968),[7] "hundreds of event studies have been conducted in the legal, financial economics, and accounting literatures... Event studies are tests of market efficiency. They test the impact, speed, and unbiasedness of the market's reaction to an event. In an efficient capital market, a security's price reaction to an event is expected to be immediate and subsequent price movement is expected to be unrelated to the event-period reaction or its prior period return," as pointed out by Kothari (2001). Kothari and Warner (2007) also point out that "event study tests are joint tests of whether abnormal returns are zero and of whether the assumed model of expected returns (i.e., the CAPM, market model, etc.) is correct."

In this paper, I do *not* ascribe any directional component to any potentially material event,[8] because it is impossible to objectively determine the market's perception prior to any potentially material event and to determine whether a particular potentially material event was better than expected news, worse than expected news, or even a surprise at all. Colloquially speaking: Good news or bad news is not the relevant question here; and reaction, overreaction, correction, overcorrection, bounceback, etc. should, in an efficient market, be out of the system within a few days after a potentially material event. I do the same analyses using the CAPM and MM separately.

For each stock, for each potentially material event at time t, I calculate the cumulative abnormal return for the $[(T-1 \text{ week}) \leq t \leq (T+1 \text{ week})]$ window, thus allowing for the possibility that material information could leak out about a week before the potentially material event and that it takes up to a week after the potentially material event for the unanticipated information to have its

[7]See Ball (2017) for a retrospective by one of the cofounders of event studies describing the contributions of the other cofounders. See Brown and Warner (1985) for a summary of event studies using daily data.

[8]If the event-time is at or after 4 PM, I consider the next day as the effective day.

impact on stock returns. From this cumulative abnormal return, I subtract the cumulative abnormal return for the $[(T - 1 \text{ week}) \leq t \leq (T + 4 \text{ days})]$ window in order to *calculate the cumulative abnormal return for the stock over the last three days of a week's window following the potentially material event for that stock* — this, as discussed above, is closer to zero as the closer a market is to efficiency. **For robustness, I have run the analyses for 2-day and 4-day windows and the results stay qualitatively the same.**

Depending on how one determines a potentially material event,[9] I have two separate paths of research on event studies. For the first path, I rely upon the marking of an event as a "Key Development" by Capital IQ, a service of Standard & Poor's, to study as a potentially material event. For each stock-year, I calculate the average, for that stock-year, of the absolute value of the cumulative abnormal return for the security over the last three[10] days of a week's window following each such Key Development for that stock in that year – I call this the *Key Developments Abnormal Response (KDAR) for that stock-year and use it as a negative measure of the efficiency of the market for the stock in that year.* As mentioned earlier, I do not try to ascribe any directional component to any Key Development, because it is impossible to objectively determine the market's perceptions immediately prior to the relevant Key Development and, therefore, to determine whether a particular Key Development was better than expected news, worse than expected news, or just expected. Colloquially speaking: Good news or bad news is not the relevant question here; and reaction, overreaction, correction, overcorrection, bounceback, etc. should, in an efficient market, be out of the system within a few days after a potentially material event.

The second path studies the impact of earnings announcements on security prices and has a long and distinguished history; see, for instance, Ball and Brown

[9] If the event-time is at or after 4 PM, I consider the next day as the effective day.

[10] To test for robustness, I redid all the analyses using two- and four-day windows — the results remain very similar.

(1968), Collins and Kothari (1989), and Kothari and Warner (2007). For each stock-year, I calculate the average, for that stock-year, of the absolute value of the cumulative abnormal return for the security over the last three[11] days of a week's window following each such earnings announcement for that stock-year — I call this the *Earnings Announcements Abnormal Response (EAAR) for that stock-year and use it as a negative measure of the efficiency of the market for the stock in that year*. Please note that the actual announced EPS or its deviation from "consensus" forecasts do not enter into my calculations; for a number of reasons. Any estimate of the market's "consensus" prediction of the EPS at the point of an EPS announcement, whether it is by using mean/medians of analyst forecasts, or by using valuation models, is sensitive to the methodology used to estimate the market's perception and the deviation from it — see, for instance, Chiang, Dai, Fan, Hong and Tu (2016) for detailed descriptions of biases in the calculation of deviation from consensus. See Kothari, So and Verdi (2016) for a comprehensive survey of the literature on quality, bias, and predictability of earnings forecasts. The impact of analyst incentives on analyst forecasts is beyond the scope of this paper. It is impossible to measure how much information about an actual earnings announcement leaked out before the actual announcement, and, therefore, to measure the "surprise" component of the earnings announcement.

3 Arbitrage Costs

It is well-understood that arbitrage activity, which is primarily responsible for correction of mispricing of a security, renders efficient the market for that security — see, for example, Harris (2003)[12] and Grossman and Stiglitz (1980).

[11] To test for robustness, I redid all the analyses using two- and four-day windows — the results remain very similar.

[12] Chapters 10 and 17, in particular.

The literature recognizes two types of arbitrage costs: transaction costs (including bid-ask spread, commissions and search costs) and holding costs — see, for example, Pontiff (2006) — if the transaction cost or holding cost relevant to a stock is higher, everything else remaining the same, the market for that stock is likely to be less efficient. Thus, arbitrage risk and idiosyncratic risk are negative measures of the efficiency of the market for a stock, and I estimate them separately based on CAPM and MM; see Section 3 for details. Consider an arbitrageur whose information suggests that a stock is underpriced; the arbitrageur will then "go long" on that stock in order to obtain arbitrage profits by selling the stock at a later date. However, the arbitrageur will also manage the risk of holding the stock by hedging. *The risk of this optimal arbitrage portfolio is the arbitrage risk of the stock and is a negative measure of market efficiency* — see Bhattacharya and O'Brien (2015) and Wurgler and Zhuravskaya (2002) for details.

Idiosyncratic risk is defined as the standard deviation of residuals of the standard Capital Asset Pricing Model (CAPM) or the Market Model (MM). Pontiff (2006) concludes that "idiosyncratic risk is the single largest cost faced by arbitrageurs" and "idiosyncratic risk is the single largest barrier to arbitrage," and therefore, *idiosyncratic risk is a negative measure of market efficiency.*[13]

4 Determinants of Market Efficiency

4.1 Trading Volume

An investor's subjective value of an asset is the price at which the investor is indifferent between the *status quo* and a trade. Note that valuations for an asset can differ across investors for two primary reasons: private value, which could

[13] Morck, Yeung and Yu (2000) interpret idiosyncratic risk as a decreasing function of "synchronicity" and "conjecture" that "synchronicity" is negatively related to "processors of economic information."

be dictated by, for instance, the investor's liquidity constraints, and private information, which could be determined by, for instance, the investor's cost of gathering information to evaluate the security. In order to appreciate how trading volume impacts market efficiency, we need to understand why a trade occurs. In particular, as Wang (1994) says, "investors trade among themselves because they are different. ... The greater the information asymmetry, the larger the abnormal trading volume when public news arrives."[14] Beaver (1968) refers to the "economist's notion that volume reflects a lack of consensus regarding the price." For example, one would observe higher trading volume in the market for an asset if and only if transaction costs (including bid-ask spread, commissions and search costs) are low relative to dispersion in investor valuations,[15] and low-valuers hold more securities than high-valuers, and short sales costs are low relative to dispersion in investor valuations. Conversely, we would observe low volume in the market for an asset if and only if transaction costs (including bid-ask spread, commissions and search costs) are high relative to dispersion in investor valuations, and high-valuers hold more assets than low-valuers, and short sales costs are high relative to dispersion in valuations. It is often the case that the holders of assets have higher valuations because of a *status quo bias* or for other private value reasons.

Consider the following example of a homeowner with a mortgage equal to $\$3,000,000$. To sell the house, the homeowner will have to pay a 6% commission, and can afford to pay at most $\$100,000$ to be clear of the mortgage; i.e., to have to pay more than $\$100,000$ out of pocket, (s)he will be better off not selling the house. Let the minimum price that the homeowner will accept be p. Then, the homeowner will get $p - 6\%\,(p)$, and so will be able to pay $p - 6\%\,(p) + \$100,000$

[14] See also Karpoff (1987), O'Hara (1997), Stickel and Verrecchia (2004), Blume, Easley and O'Hara (1994), and Goldstein, Hotchkiss and Sirri (2007).

[15] With high dispersion in investor valuations, we can have a high trading volume, much of which can be due to noise trading. A higher dispersion in investor valuations does not necessarily add to the efficiency of a market.

to clear the mortgage; i.e., $p - 6\%\,(p) + \$100,000 = \$3,000,000$ solves for $p = \$3,085,100$, which is therefore, a lower bound on the homeowner's valuation.[16] This lower bound depends, for instance, positively on transaction cost and negatively on the cash or cash-equivalents available to the homeowner. Therefore, the likelihood of a trade depends negatively on transaction cost (commission in this case) and positively on the homeowner's solvency. This is an example of a situation where the lowest price at which the owner will sell is often higher than the highest price at which a potential buyer will buy, and in such a case, since short sales costs are very high, there will be no trade.[17]

Higher trading volume will therefore be observed with lower short sales costs, lower transaction costs, higher dispersion of investor valuations, and/or higher likelihood that an investor holds the asset. However, for instance, there is no reason that higher dispersion in investor valuations leads to higher market efficiency — in fact, I show an example in Subsection 4.1.1 where there is no dispersion in valuations and the market is perfectly semistrong form efficient but with no trades — and therefore, the impact on market efficiency of normalized trading volume, everything else remaining the same, is fundamentally an empirical question, and therefore, the impact of short sales costs & constraints on market efficiency, everything else remaining the same, is also an empirical question.

4.1.1 Simple Model of Trading Volume

Let the set of all investors (and potential investors) at time t be N_t, and let for $i \in N_t$, the valuation of investor i be $v_{i,t}$ per unit of the security and let the holding of investor i at the beginning of time t be $h_{i,t}$ (which could be positive, zero or negative, the last for short positions). Let $\Theta_{i \longrightarrow j}$ denote the number of

[16] As mentioned earlier, an investor's subjective value of an asset is the price at which the investor is indifferent between the *status quo* and a trade.

[17] Short sales costs are often prohibitive for certain assets such as real estate.

shares sold by investor i to investor j in period t; obviously, $\Theta_{j\longrightarrow i} = -\Theta_{i\longrightarrow j}$. Let the mean transaction costs (e.g., bid-ask spread, commissions, search and matching costs) per unit of a trade at time t be τ_t, and let the mean short sales costs per unit of a trade at time t be ψ_t. Let the total cost of a trade $\Theta_{i\longrightarrow j}$ be denoted $C(\Theta_{i\longrightarrow j})$.

Condition 1 ***(Constant Average Valuation & Trading Cost Condition)*** *Per-unit valuation $v_{i,t} \geq 0$ is constant $\forall i, t$, and per-unit transaction cost $\tau_t \geq 0$ and per-unit short sales costs $\psi_t \geq 0$ are constant, $\forall t$.*

Remark 2 *Optimality will require that a trade will occur only from an investor from a lower valuation to an investor with a higher valuation. In other words, $v_i > v_j \Longrightarrow \Theta_{i\longrightarrow j} \leq 0$, or equivalently, $\Theta_{i\longrightarrow j} > 0 \Longrightarrow v_i \leq v_j$. If any of these costs is positive, these weak inequalities can be replaced by strong inequalities, which can hold also when trade is set as the default.*

Remark 3 *Any trade will incur the transaction costs at τ_t per unit, and if any portion of it is a short sale, the short sale portion will incur an additional ψ_t per unit. In other words, let $v_i < v_j$, then if $\Theta_{i\longrightarrow j} \leq h_{i,t}$ then $C(\Theta_{i\longrightarrow j}) = \tau_t \Theta_{i\longrightarrow j}$, and if $\Theta_{i\longrightarrow j} > h_{i,t}$ then $C(\Theta_{i\longrightarrow j}) = \tau_t h_{i,t} + (\tau_t + \psi_t)(\Theta_{i\longrightarrow j} - h_{i,t})$.*

Definition 4 *Consider random variable sequences $Z_t = (Z_{1,t}, Z_{2,t}, ...)$ and $Z' = \left(Z'_{1,t}, Z'_{2,t}, ...\right)$ such that $\boldsymbol{P}(|Z_i - Z_j| > \kappa) > \boldsymbol{P}\left(\left|Z'_i - Z'_j\right| > \kappa\right), \forall \kappa > 0, \forall i, j \in N$. Then Z is more **dispersed in tail-probabilities of differences,** or simply, more **dispersed,** than Z'.*

Proposition 5 *If $v_{i,t} = v^*_t + \varepsilon_{i,t}$ where $(\varepsilon_{i,t})_{i\in N}$ are iid $\mathcal{N}\left(0, \sigma^2_t\right)$, and $v'_{i,t} = v'^*_t + \varepsilon'_{i,t}$ where $\left(\varepsilon'_{i,t}\right)_{i\in N}$ are iid $\mathcal{N}\left(0, \sigma'^2_t\right)$, then the valuation profile $v_t = (v_{1,t}, v_{2,t}, ...)$ is more dispersed than $v'_t = \left(v'_{1,t}, v'_{2,t}, ...\right)$ if and only if $\sigma^2_t > \sigma'^2_t$.*

Condition 6 ***(Independence Condition)*** $v_{i,t}$ *is stochastically independent of* $h_{i,t} \forall i$*. In other words, an investor's valuation of the security is stochastically independent of her/his holding of the security at the beginning of the period.*

Condition 7 ***(A Priori Identical Investors Condition)*** *Investor holding probability* $\boldsymbol{P}(h_{i,t} > 0)$ *is, a priori, identical across investors:* $\boldsymbol{P}(h_{i,t} > 0) = \pi_t, \forall i$*.*

Lemma 8 *Under Conditions 1, 6 and 7,* $\forall i, j$*,*

1. $\boldsymbol{P}(\Theta_{i \longrightarrow j} \neq 0)$ *is, ceteris paribus, higher if the valuation profile is more dispersed.*
2. $\boldsymbol{P}(\Theta_{i \longrightarrow j} \neq 0)$ *is, ceteris paribus, increasing in* π_t*.*
3. $\boldsymbol{P}(\Theta_{i \longrightarrow j} \neq 0)$ *is, ceteris paribus, decreasing in* τ_t*.*
4. $\boldsymbol{P}(\Theta_{i \longrightarrow j} \neq 0)$ *is, ceteris paribus, decreasing in* ψ_t*.*

In other words, under the Constant Average Cost Condition, Independence Condition and *A Priori* Identical Investors Condition, the probability of a trade between investors i and j is, *ceteris paribus,* higher

1. if the valuation profile is more dispersed — as Wang (1994) says, "investors trade among themselves because they are different," and Beaver (1968) refers to the "economist's notion that volume reflects a lack of consensus regarding the price."
2. if the probability of an investor having a positive holding in the security is higher.
3. if transaction cost is lower.
4. if short sales costs are lower.

Proof. Case 1. $v_{i,t} > v_{j,t}$. Then $\Theta_{i \longrightarrow j} \leq 0$. In other words, there will be no trade such that the investor with the higher valuation sells to the investor with the lower valuation.

Case 1a. $v_{i,t} - v_{j,t} > \tau_t + \psi_t$. Then, $\Theta_{j \longrightarrow i} > 0$. In other words, the investor with the lower valuation will sell (could be a short sale) to the investor with the higher valuation.

Case 1b. $\tau_t \leq v_{i,t} - v_{j,t} < \tau_t + \psi_t$. Then, $\Theta_{j \longrightarrow i} > 0 \Longleftrightarrow h_{j,t} > 0$. In other words, the investor with the lower valuation will sell to the investor with the higher valuation if and only if the trade is not a short sale.

Case 1c. $v_{i,t} - v_{j,t} < \tau_t$. Then, $\Theta_{j \longrightarrow i} = 0$. In other words, there will be no trade since the difference in valuations is smaller than the transaction cost.

Case 2 ($v_{i,t} < v_{j,t}$) is a mirror image of Case 1. Therefore, the probability of a trade between investors i and j = $\boldsymbol{P}\left(\Theta_{i \longrightarrow j} \neq 0\right) = \boldsymbol{P}\left(\{\Theta_{i \longrightarrow j} > 0\} \cup \{\Theta_{j \longrightarrow i} > 0\}\right) =$ $\boldsymbol{P}\left(v_{i,t} - v_{j,t} > \tau_t + \psi_t\right) + \boldsymbol{P}\left(v_{j,t} - v_{i,t} > \tau_t + \psi_t\right) + \boldsymbol{P}\left(\tau_t \leq v_{i,t} - v_{j,t} < \tau_t + \psi_t\right) \boldsymbol{P}\left(h_j > 0\right) +$ $\boldsymbol{P}\left(\tau_t \leq v_{j,t} - v_{i,t} < \tau_t + \psi_t\right) \boldsymbol{P}\left(h_i > 0\right)$ (from Conditions 1 and 6) $= \boldsymbol{P}\left(|v_{i,t} - v_{j,t}| > \tau_t + \psi_t\right) +$ $\pi_t \boldsymbol{P}\left(\tau_t \leq |v_{i,t} - v_{j,t}| < \tau_t + \psi_t\right)$ (from Condition 7) $= \pi_t \boldsymbol{P}\left(|v_{i,t} - v_{j,t}| > \tau_t\right) +$ $(1 - \pi_t) \boldsymbol{P}\left(|v_{i,t} - v_{j,t}| > \tau_t + \psi_t\right)$ is, *ceteris paribus,* higher if the valuation profile is more dispersed, increasing in π_t, and decreasing in τ_t and ψ_t. Therefore, a higher trading volume will be observed with lower short sales costs, lower transaction costs, higher dispersion of investor valuations, and/or higher likelihood that an investor holds the asset. $\blacksquare$

It is not the case that higher dispersion would unambiguously facilitate market efficiency. Consider what would happen in the limiting case where there is practically zero dispersion in investor valuations. Then, every investor would have identical valuations v_t^*, and the market would be perfectly semistrong form efficient with a notional equilibrium price $p_t = v_t^*$ that would reflect all the information available to the investors, albeit with no trades. In this simple model, if dispersion were to increase, the market would become less efficient. To see this more clearly, consider the special case outlined in Remark 5: $v_{i,t} = v_t^* + \varepsilon_{i,t}$ where $(\varepsilon_{i,t})_{i\in N}$ are iid $\mathcal{N}\left(0, \sigma_t^2\right)$, and $v'_{i,t} = v_t'^* + \varepsilon'_{i,t}$ where $\left(\varepsilon'_{i,t}\right)_{i\in N}$ are iid $\mathcal{N}\left(0, \sigma_t'^2\right)$, then the valuation profile $v_t = (v_{1,t}, v_{2,t}, ...)$ is more dispersed than $v'_t = \left(v'_{1,t}, v'_{2,t}, ...\right)$ if and only if $\sigma_t^2 > \sigma_t'^2$. By the previous discussion, we see that the limiting case of $\sigma_t'^2 = 0$ would correspond to a perfect semistrong efficient market with a notional equilibrium price $p_t = v_t^*$ that would reflect all the information available to the investors, albeit with no trades. In this simple model, if we had a higher $\sigma_t^2 > \sigma_t'^2$, *ceteris paribus,* the market would become less efficient.

In conclusion, the impact on market efficiency of normalized trading volume, everything else remaining the same, is fundamentally an empirical question, and the impact of short sales costs & constraints on market efficiency, everything else remaining the same, is also an empirical question.

4.2 Number of Market Makers

From the above discussion, we can see that the demand for market making services is an increasing function of trading volume, for instance, through higher dispersion of the valuation profile. As a corollary, everything else remaining the same,[18] a firm is more likely to enter (or not exit) the market for market making services if there is higher trading volume (and thus, higher market making profits), for instance, through a higher dispersion of the valuation profile for that security. However, the higher the number of market makers, competition for trades would put downward pressure on the transaction costs τ_t, and economies of scale will determine the equilibrium impact on the price of market making services (τ_t).[19] Therefore, the direction of the impact of the number of

[18] In particular, keeping constant other incentives of investment banks, such as profits from proprietary trading.

[19] See Bhattacharya (2017), for instance.

(Nasdaq) market makers for a security on the efficiency of the market for that security can only be determined empirically.

5 Description of Data

Stock-days[20] that have positive closing price and positive shares outstanding[21] and other restrictions[22] from CRSP: date, closing price, return, shares outstanding, trading volume, closing bid and ask, exchange membership, number of market makers, SIC Code, value-weighted market return of all publicly traded U.S. stocks,[23] and equally-weighted market return of all publicly traded U.S. stocks.[24] And had data on the following variables: analyst forecasts (from I/B/E/S), number of securities analysts (from I/B/E/S), earnings announcements (from I/B/E/S), institutional holdings (from Thomson Reuters), short interest (from Compustat), Key Developments (from Capital IQ), Compustat-CRSP Merged Database.[25]

- https://drive.google.com/open?id=1t9UNjxSSv85x3ETZ5dez77lKXXDIHArS

[20] For a stock to be in my sample for a particular calendar year, I require that the stock have traded for at least half the trading days in that calendar year. In order to maintain the integrity of the linkages between the different sources of data I rely upon, I treat a missing observation as a missing observation — in particular, with number of analysts, short interest, institutional ownership, and number of market makers — not as a zero. Appendix 2 summarizes the relationship between the various factors used in this paper — I use unweighted averages across time (such as annual average of daily data).

[21] For a small number of stock-days, I found multiple entries when sorted by ticker, CUSIP, and date — for these stock-days, I calculate weighted averages using trading volume as weights.

[22] If shrcd in (10,11); if ret notin (-44.0,-55.0,-66.0,-77.0,-88.0,-99.0); if dlret notin (-55.0,-66.0,-88.0,-99.0); if dlretx notin (-55.0,-66.0,-88.0,-99.0); if dclrdt ne 0; if trtscd ne 0; if nmsind ne 0; if vol ne -99.0; if prc > 0; if bid > 0; if ask > 0; if vol > 0; if shrout > 0; if ticker ne "?"; if ticker ne ""; if cusip ne ""; if substr(ticker,1,4) ne "ZZZZ"; if substr(cusip,7,1) in ("0","1","2","3","4","5","6","7","8","9"); if substr(cusip,8,1) in ("0","1","2","3","4","5","6","7","8","9").

[23] "VWRETD indices contain ... the daily ... returns, including all distributions, on a value-weighted market portfolio (excluding American Depository Receipts (ADRs))." (CRSP information)

[24] "EWRETD indices contain ... daily ... returns, including all distributions, on an equally-weighted market portfolio (including ADRs)." (CRSP information)

[25] Compustat (including Capital IQ) data sources rely upon the S&P identification of GVKEY. In this paper, I rely upon the mapping of GVKEY to TICKER used by CRSP to match its data to Compustat.

- https://drive.google.com/open?id=1IcTYOQRVCfYjZB94s3haHyMGpzW4cAuj
- https://drive.google.com/open?id=1DsHQR7VyoimRYmF-b0agvHlOzyJjyD_5
- https://drive.google.com/open?id=1VyQGls9Gtomo52Fvxp5K1SMYPCsMtw4B
- https://drive.google.com/file/d/1cAYFmJcJsLiYebx8YkYQNftyNP0VdsRA/view?usp=sharing
- https://drive.google.com/file/d/10rkMaWUyq3q2MdcaxXbRZqo17OlZrKuI/view?usp=sharing

6 Measures of Market Efficiency

In this paper, I use eight different and objective metrics described earlier as *separate negative measures* of the efficiency of the market for a stock.

6.1 Based on event studies

1. Key Developments Abnormal Response (KDAR) of the security, estimated through the Capital Asset Pricing Model (CAPM)
2. KDAR of the security, estimated through the Market Model (MM)[26]
3. Earnings Announcements Abnormal Response (EAAR) of the security, estimated through CAPM
4. EAAR of the security, estimated through MM

6.2 Based on economic cost considerations

1. Arbitrage Risk (AR) of the security, estimated through CAPM
2. Arbitrage Risk (AR) of the security, estimated through MM
3. Idiosyncratic Risk (IR) of the security, estimated through CAPM

[26] See Brown and Warner (1985) for a discussion of the similarity of the econometric results of MM and CAPM.

4. Idiosyncratic Risk (IR) of the security, estimated through MM

7 Econometric Methodology

7.1 Potential Factors of Market Efficiency

Appendix 2 provides summary statistics and correlation coefficients of the following descriptors of interest: volume, market cap, bid-ask spread,, short interest, number of analysts, standard deviation of analyst forecasts, and number of market makers (Nasdaq stocks only).

In this paper, I incorporate the Fama-French Factors (market cap, book-to-market equity ratio, price-to-earnings ratio, and leverage) and institutional ownership percentage in the following two ways: *Fixed Effects* (separately for each Fama-French Factor and institutional ownership percentage, fixed effects for the deciles for that factor), and *Separate Regressions:* (separate regression system for each decile for each Fama-French Factor and institutional ownership percentage).

7.2 Testing of One-Tailed Hypotheses

For the impact on market efficiency of short sales costs & constraints, I do the following **one-tailed test :** The null hypothesis $H_0 : \beta_{MEff,SSCC} \left(= \frac{\partial(\text{Market Efficiency})}{\partial(\text{Short Sales Costs \& Constraints})} \right) \geq \mathbf{0}$ versus the alternate hypothesis $H_1 : \beta_{MEff,SSCC} < \mathbf{0}$.

The parameter of interest, $\beta_{MEff,SSCC}$, is estimated by multiplying the 3SLS estimates of 1) the coefficient of volume in the market efficiency equation and 2) the coefficient of short sales costs & constraints in the volume equation — i.e., by multiplying estimates of parameters $\alpha_{SSC,i}$ and $\gamma_{Vol,i}$ (each produced by, for instance, PROC MODEL in SAS). The *p-value* of this one-tailed test above is obtained by adjusting by 1/2 the *p-value* (reported by SAS in PROC MODEL, for instance) for the χ^2-statistic corresponding to the (two-tailed) Wald Test of the (non-linear) null hypothesis $(\alpha_{SSC,i})\left(\gamma_{Vol,i}\right) = 0$ versus the alternate (two-tailed) hypothesis $(\alpha_{SSC,i})\left(\gamma_{Vol,i}\right) \neq 0$.

For the impact of normalized volume on market efficiency, I do the following **one-tailed test:** The null hypothesis $H_0 : \gamma_{Vol}\left(= \frac{\partial(\text{Market Efficiency})}{\partial(\text{Normalized Volume})}\right) \leq \mathbf{0}$ versus the alternate hypothesis $H_1 : \gamma_{Vol} > \mathbf{0}$.

Here, the 3SLS estimate of the parameter of interest, γ_{Vol}, is provided directly (by, for instance, PROC MODEL in SAS). The *p-value* of this one-tailed test above is obtained by adjusting by 1/2 the *p-value* reported (by SAS, for instance, in PROC MODEL) for the t-statistic corresponding to the two-tailed t test of the (linear) null hypothesis $\gamma_{Vol} = 0$ versus the alternate (two-tailed, linear) hypothesis $\gamma_{Vol} \neq 0$.

7.3 Instruments

In a system of equations $Y = X\beta + \varepsilon$, for a vector W to be an instrument, it is required that **Strong First Stage:** $Cov\,(W, X) \neq 0$, and **Exclusion Restriction:** $Cov\,(W, \varepsilon) = 0$.[27] As Wooldridge (2010) points out, most of the problems with using instrumental variables arise in small samples, which, obviously, is not the case in this paper. Wooldridge (2010) points out that "*asymptotically*, we can do no worse, and can often do better, using a larger set of *valid* instruments" [emphases added]. Given these two critical qualifications, I consider in this paper the following instrumentation categories, **together and also separately**, to ensure that my conclusions are not sensitive to particular information assumptions.[28] In panel data econometrics, there are two popular

[27] In other words, the unexplained part ("residual") ε of the regression needs to be uncorrelated with the instrument — this *does not* require that the regressand Y itself be uncorrelated with the instrument.

[28] I am very grateful to Stefan Nagel for emphasizing the need for a robust identification strategy. I explain the reasons for the instrumentation strategy for each of the endoge-

categories of autochthonous instruments.

1. Instrument a variable $X_{i,t}$ by combination(s) of $X_{j,t}, j \neq i$. Cross sectional variables such as corresponding variables for other geographies have been used as instruments by, for instance, Hausman, Leonard and Zona (1994) and the validity of such instrumentation depends on the assumption that the corresponding cross sectional variables are correlated with the instrumented variable because of cost (or other) commonalities in the time period of interest but are not influenced by the idiosyncrasies of the geography of interest in the particular time period.

2. Instrument a variable $X_{i,t}$ by its lag(s) $X_{i,t-1}, X_{i,t-2}, ...$ The use of lags as instruments depends on two implicit assumptions: a) lack of serial correlation, see, for example, Greene (2018), and b) weak rational expectations, see, for example, Hansen and Singleton (1982).[29]

7.4 Structural System

I estimate the following structural system at the stock-year level,[30] based on the theory described in Section 4:

$$\begin{aligned} Vol_{i,t} &= \alpha_i + \alpha_{SSC,i} SSCC_{i,t} + \alpha_{Disp,i} Disp_{i,t} + \alpha_{Hold,i} Hold_{i,t} \\ &\quad + \alpha_{BA,i} BASpr_{i,t} + \alpha_{MM,i} MM_{i,t} + \varepsilon_{i,t} \end{aligned}$$

$$MEff_{i,t} = \gamma_i + \gamma_{Vol,i} Vol_{i,t} + \gamma_{NAn,i} NAn_{i,t} + \delta_{i,t}$$

nous/proxy variables in Subsection 7.4.

[29] Wintoki, Linck and Netter (2012) and Roberts and Whited (2011) show that the traditional statistical tests for the use of lagged variables in panel estimation are not particularly useful, so I do not use them in this paper.

[30] Obviously, the number of market makers is not a regressor for non-Nasdaq stocks.

1. **Normalized trading volume** ($Vol_{i,t}$) — annual mean of (daily trading volume)/(daily shares outstanding) — of the stock for the current calendar year is a function of

   (a) **Short sales costs and constraints** ($SSC_{i,t}$). I use the annual mean of the short interest ($SI_{i,t}$) in the stock, normalized by annual mean shares outstanding, as a negative proxy for short sales costs & constraints for the stock.[31] This is based on the *Law of Demand* hypothesis that, *ceteris paribus*, the short interest for a stock (the quantity demanded) is a decreasing function of the short sales costs & constraints (price), since, as Lamont and Thaler (2003) show, the supply is fairly constant as a function of price. I consider as instrument the mean, across all stocks, of all monthly short interest $\left(\overline{SI_t}\right)$ because the short interest for the stock this year is likely to be correlated with short interest of all stocks in the current calendar year due to regulatory and/or market reasons with minimal correlation with the disturbance term in the equation connecting normalized short interest in the stock in the current period ("quantity" of short interest) with short sales costs & constraints ("price" of short interest) of the stock in the current period. I also consider the once-lagged short interest ($SI_{i,t-1}$) as an instrument because the short interest for the stock during the previous calendar year is likely to be correlated with short interest of the stock in the current calendar year due to regulatory and/or market reasons without being correlated with the disturbance term in the equation connecting normalized short interest in the stock in the current period: ("quantity" of short in-

[31] I do not have any data on short sales costs. See Ofek, Richardson and Whitelaw (2004) and Cremers and Weinbaum (2010), for instance, on papers that use proprietary data on rebate rates and spreads.

terest) with short sales costs & constraints ("price" of short interest) of the stock in the current period.

(b) **Dispersion of investor valuations** ($Disp_{i,t}$). As a positive proxy for the dispersion of investor valuations, I use the annual standard deviation of analyst forecasts of earnings per share for the firm with "forecast period end date" in the current calendar year ($sd\left(AnF_{i,t}\right)$). It is based on the assumption that the range of security analysts reflects the range of investor sentiment. As instrument, I consider the mean across all firms of the annual mean of the quarterly standard deviation of analyst forecasts of earnings per share for the firm with "forecast period end date" in the same calendar year $\left(\overline{sd\left(AnF_{i,t}\right)}\right)$. I do so since information flows are likely to be somewhat commonly affected across different stocks in the same calendar year while being only minimally correlated with the disturbance term in the equation connecting dispersion of investor valuations with dispersion of investor valuations in the same stock and same calendar year. As instrument, I also consider the annual mean of the quarterly standard deviation of analyst forecasts of earnings per share for the firm with "forecast period end date" in the previous calendar year ($sd\left(AnF_{i,t-1}\right)$) since information flows are likely to be contemporaneously affected without being correlated with the disturbance term in the equation connecting dispersion of investor valuations in the stock in the current calendar year with standard deviation of analyst forecasts of the stock in the previous calendar year.

(c) **Investor holding probability** ($Hold_{i,t}$). I use annual mean of natural logarithm of daily shares outstanding ($\ln SO_{i,t}$) as a positive proxy of the investor holding probability. Under the *A Priori* Identical Investors Condition, this is a relatively constant multiple of the investor holding probability, so I do not need to instrument.

(d) **Transaction costs**, which are functions of a) **Relative bid-ask spread** ($BASpr_{i,t}$), and b) **Number of marker makers** ($MM_{i,t}$). The former is measured by annual mean of daily relative bid-ask spread where daily relative bid-ask spread = absolute(daily closing best ask – daily closing best bid)/(mid-point of daily closing best bid and best ask). It is obviously endogenous. I consider as instrument the mean across all stocks of the annual mean of daily relative bid-ask spread for the stock for the same calendar year $\left(\overline{BASpr_t}\right)$ because the bid-ask spread of other stocks during the current calendar year is likely to be correlated with bid-ask spread of the relevant stock during the current calendar year due to cost and/or market reasons, but with minimal correlation with the disturbance term in the trading volume equation for the stock in the current period. I also consider as instrument the annual mean of daily relative bid-ask spread for the stock for the previous calendar year ($BASpr_{i,t-1}$) because the bid-ask spread of the stock during the previous calendar year is likely to be correlated with bid-ask spread of the relevant stock during the current calendar year due to cost and/or market reasons, without being correlated with the disturbance term in the trading volume equation for the stock in the current period. The number of marker makers ($MM_{i,t}$)[32] is measured by the mean over the calendar year of the number of market makers for the stock on each day. It is also endogenous, and I consider as instrument the mean across all stocks of the mean of the number of market makers for the stock in the current

[32] For obvious reasons, I do not use this regressor when I do the regressions for non-Nasdaq stocks.

calendar year $\left(\overline{MM_t}\right)$ because the number of market makers for other stocks in a calendar year is likely to be correlated with the number of market makers for the relevant stock in the current calendar year due to cost and/or market reasons but with minimal correlation with the disturbance term in the trading volume equation for the stock in the current calendar year. I also consider as instrument the mean of number of market makers for the stock in the previous calendar year $(MM_{i,t-1})$, because the number of market makers for a stock in the previous calendar year is likely to be correlated with the number of market makers for the stock in the current calendar year due to cost and/or market reasons without being correlated with the disturbance term in the trading volume equation for the stock in the current calendar year.

2. **Market efficiency** $(MEff_{i,t})$ — I use the **negative** of each different metric as a separate **positive measure** of the efficiency of the market for a stock; it is the regressand and hence, there is no need for instrumentation. It is a function of

   (a) **Normalized trading volume** $(Vol_{i,t})$ — annual mean of (daily trading volume)/(daily shares outstanding) is endogenous and I consider as instrument the mean across all stocks of the annual mean of (daily trading volume)/(shares outstanding) of the stock during that year $\left(\overline{Vol_t}\right)$ because trading volume of other stocks during the current calendar year is likely to be correlated with trading volume of the relevant stock in the current calendar year due to regulatory and/or market reasons but with only minimal correlation with the disturbance term in the market efficiency equation for the stock in the current calendar year. I also consider as instrument the annual mean of (daily trading volume)/(shares outstanding) corresponding to the stock for the previous calendar year ($Vol_{i,t-1}$) because trading volume of the stock during the previous calendar year is likely to be correlated with trading volume of the stock in the current calendar year due to regulatory and/or market reasons without being correlated with the disturbance term in the market efficiency equation for the stock in the current calendar year.

(b) **Number of security analysts** ($NAn_{i,t}$) is measured by the annual average of the quarterly number of analysts who made EPS forecasts for the firm with "forecast period end date" in the current calendar year. It is also endogenous, and I consider as instrument the mean across all firms of the number of analysts who made EPS forecasts for the firm with "forecast period end date" in the same calendar year $\left(\overline{NAn_t}\right)$ because the numbers of analysts for other firms in the same calendar year are likely to be correlated with the number of analysts for the firm in the current calendar year due to cost and/or market reasons but with minimal correlation with the disturbance term in the market efficiency equation for the stock in the current calendar year. I also consider as instrument the average of the number of analysts who made EPS forecasts for the firm with "forecast period end date" in the previous calendar year ($NAn_{i,t-1}$) because the number of analysts for the firm in the previous calendar year is likely to be correlated with the number of analysts for the firm in the current calendar year due to cost and/or market reasons without being correlated with the disturbance term in the market efficiency equation for the stock in the current calendar year.

8 Empirical Results and Their Implications

I perform each of my analyses, separately for all Nasdaq stocks and all non-Nasdaq U.S. stocks, for 1996 through 2016 (full calendar years for which I have complete data).[33] In this paper, I incorporate the Fama-French Factors (market cap, leverage, book-to-market equity ratio, and price-to-earnings ratio) and institutional ownership percentage in the following two ways: a) *Fixed Effects:* Separately for each Fama-French Factor and institutional ownership percentage, fixed effects for the deciles for that factor, and b) *Separate Regressions:* Separate regression system for each decile for each Fama-French Factor and institutional ownership percentage.

The empirical results are robust across the different measures of market efficiency, across alternate specifications and instrumentations, and across the Fama-French Factors and institutional ownership percentage. Unless noted otherwise, all significances reported in this paper are at 5%, and can be summarized as follows: Separately for Nasdaq and non-Nasdaq stocks,[34] the impact of short sales costs & constraints on market efficiency is *not* negative, and that of normalized trading volume is *not* positive.[35]

- https://drive.google.com/drive/folders/1cPV0_s1WMiPrSyZ0Ci4cMm-bXbHDVb-t?usp=sharing

[33] For obvious reasons, I do not have the number of market makers as a regressor when I run regressions for non-Nasdaq stocks.

[34] **As an additional set of analyses**, I also took the naive approach and estimated the corresponding one-equation reduced form, but as with this paper, correcting for endogeneity and errors-in-variables, and incorporating the Fama-French (1992) Factors and institutional ownership (by using fixed effects): $MEff_{i,t} = \gamma_i + \lambda_{SSC,i}SSC_{i,t} + \lambda_{Vol,i}Vol_{i,t} + \lambda_{Disp,i}Disp_{i,t} + \lambda_{Hold,i}Hold_{i,t} + \lambda_{BA,i}BA_{i,t} + \lambda_{MM,i}MM_{i,t} + \lambda_{MCap,i}MCap_{i,t} + \lambda_{NAn,i}NAn_{i,t} + \eta_{i,t}$. This set of Two-Stage Least Squares regressions yields similar conclusions about the *association* of market efficiency with a) short sales costs & constraints and b) normalized trading volume — note that I use *association* rather than *impact* in the context of interpreting one equation that is not directly based upon theory; because there is no reason based on financial economics or information economics for market efficiency to be a *direct function* of short sales costs & constraints.

[35] This was also the conclusion of Bhattacharya and O'Brien (2015) using a reduced form on arbitrage risk as a negative proxy for market efficiency.

8.1 Do Short Sales Costs & Constraints Negatively Affect Market Efficiency?

$$\mathbf{H}_0\text{: } \boldsymbol{\beta}_{MEff,SSCC} \geq \mathbf{0} \text{ vs. } \mathbf{H}_1\text{: } \boldsymbol{\beta}_{MEff,SSCC} < \mathbf{0}$$

8.1.1 Using Fama-French Factor and Institutional Ownership Deciles as Fixed Effects

	Nasdaq Stocks		Non-Nasdaq Stocks	
	Number of	Number of	Number of	Number of
Factor	**Regressions**	**Rejections**	**Regressions**	**Rejections**
Overall	120	8	120	20
Book to Market Ratio	24	0	24	4
Leverage	24	0	24	4
Market Cap	24	8	24	4
Price/Earnings Ratio	24	0	24	4
Institutional Ownership	24	0	24	4

$$\mathbf{H}_0\text{: } \boldsymbol{\beta}_{MEff,SSCC} \geq \mathbf{0} \text{ vs. } \mathbf{H}_1\text{: } \boldsymbol{\beta}_{MEff,SSCC} < \mathbf{0}$$

8.1.2 Doing Separate Regressions for Each Fama-French Factor and Institutional Ownership Decile

	Nasdaq Stocks		Non-Nasdaq Stocks	
Factor	**Number of Regressions**	**Number of Rejections**	**Number of Regressions**	**Number of Rejections**
Overall	1,164	76	1,156	147
Book to Market Ratio	240	22	240	32
Leverage	240	23	240	35
Market Cap	240	24	240	18
Price/Earnings Ratio	240	7	240	48
Institutional Ownership	204	0	196	14

8.2 Does Relative Trading Volume Positively Affect Market Efficiency?

$$\mathbf{H}_0\text{: } \boldsymbol{\beta}_{MEff,Vol} \leq \mathbf{0} \text{ vs. } \mathbf{H}_1\text{: } \boldsymbol{\beta}_{MEff,Vol} > \mathbf{0}$$

8.2.1 Using Fama-French Factor and Institutional Ownership Deciles as Fixed Effects

	Nasdaq Stocks		**Non-Nasdaq Stocks**	
	Number of	**Number of**	**Number of**	**Number of**
Factor	**Regressions**	**Rejections**	**Regressions**	**Rejections**
Overall	120	0	120	30
Book to Market Ratio	24	0	24	6
Leverage	24	0	24	6
Market Cap	24	0	24	6
Price/Earnings Ratio	24	0	24	6
Institutional Ownership	24	0	24	6

$$\mathbf{H}_0\text{: } \boldsymbol{\beta}_{MEff,Vol} \leq \mathbf{0} \text{ vs. } \mathbf{H}_1\text{: } \boldsymbol{\beta}_{MEff,Vol} > \mathbf{0}$$

8.2.2 Doing Separate Regressions for Each Fama-French Factor and Institutional Ownership Decile

	Nasdaq Stocks		Non-Nasdaq Stocks	
Factor	Number of Regressions	Number of Rejections	Number of Regressions	Number of Rejections
Overall	1,164	38	1,156	197
Book to Market Ratio	240	6	240	45
Leverage	240	12	240	58
Market Cap	240	14	240	20
Price/Earnings Ratio	240	4	240	58
Institutional Ownership	204	2	196	16

9 Conclusions and Future Research

In this paper, I utilized eight different metrics as *separate and objective negative measures* of the efficiency of the market for a stock. I developed a simple two-equation structural system. In the first equation, I modeled trading volume (and number of market makers, for Nasdaq stocks) as functions of dispersion of investor valuations, investor holding probability, transaction cost, and short sales costs. In the second equation, I modeled market efficiency as a function of normalized trading volume and number of security analysts following the stock. I showed that higher trading volume will be observed with lower short sales costs, lower transaction costs, higher dispersion of investor valuations, and/or higher likelihood that an investor holds the asset. However, for instance, there is no reason that higher dispersion in investor valuations would lead to higher market efficiency — in fact, I showed an example where there is no dispersion in valuations and the market is perfectly semistrong form efficient but with no trades — and therefore, the impact on market efficiency of normalized trading volume, everything else remaining the same, is fundamentally an empirical question, and therefore, the impact on market efficiency of short sales costs, everything else remaining the same, is also an empirical question. The relevant consequence of my two-equation system is that short sales costs & constraints impact trading volume that in turn impacts market efficiency. I used the natural logarithm of shares outstanding as a positive proxy for investor holding probability. I used normalized short interest as a negative proxy for short sales costs & constraints. I used standard deviation of analyst estimates to positively proxy for dispersion of investor valuations. I modeled bid-ask spread and number of market makers (the latter only for Nasdaq stocks) as determinants of transaction costs. I used Errors-In-Variables methods to incorporate the proxies and I used the panel nature of the data to identify appropriate instruments. I checked for robustness of my results under different identification strategies. I used Three-Stage Least Squares to estimate this structural system, and I incorporated the Fama-French Factors (market cap, leverage, book-to-market equity ratio, and price-to-earnings ratio) and institutional ownership percentage in the following two ways: *Fixed Effects:* Separately for each Fama-French Factor and institutional ownership percentage, fixed effects for the deciles for that factor, and *Separate Regressions:* Separate regression system for each decile for each Fama-French Factor and institutional ownership percentage.

I performed each of these analyses separately for all Nasdaq and all non-Nasdaq U.S. stocks (with the number of market makers an additional regressor only for the former), using daily data on all U.S.-traded stocks from CRSP, I/B/E/S, Thomson Reuters, Compustat, Capital IQ, and CRSP-Compustat Merged Database. I found that, contrary to *much* previous theoretical and empirical work, *ceteris paribus*, the efficiency of the market for the stock is not negatively impacted by short sales costs & constraints and is not positively impacted by normalized trading volume.

The economic theory and empirics of this paper can be summarized as: Separately for Nasdaq and non-Nasdaq stocks, with and without controls for Fama-French Factors and institutional ownership, normalized trading volume is negatively and significantly impacted by short sales costs & constraints, and a) volume does *not* positively and significantly impact market efficiency, and therefore, b) the impact on market efficiency of short sales costs & constraints is *not* negatively significant.

The first future research project would involve application of the theoretical and econometric methodologies of this paper to intraday data instead of daily data — such analyses are only possible now with Big Data storage, processing power, and techniques. Further along, I intend to utilize the three measures of market efficiency based on securities prices following random walks in efficient markets used by Cao, Liang, Lo and Petrasek (2018), to use the two measures of market efficiency, based on the asymmetry between positive and negative market returns used by Bris, Goetzmann and Zhu (2007), to utilize the variance ratio measures of random walk — see Campbell, Lo and MacKinlay (1997) for details, and to use the friction measures of market efficiency — see Hou and Moskowitz (2005). I also intend to explore alternative measures of market efficiency such as intraday bid-ask spreads, structural model of the relation between capital markets and industrial organization of corresponding product markets, and the framework of an expected utility-maximizing investor using different utility functions.

10 References

1. Yang Bai, Eric Chang and Jiang Wang, "Asset Prices Under Short-Sale Constraints," *MIT Working Paper*, 2006

2. Ray Ball, "Fama, Fisher, Jensen, and Roll (1969) — Retrospective Comments," in John Cochrane and Tobias Moskowitz, eds., *The Fama Portfolio*, 2017

3. Ray Ball and Philip Brown, "An Empirical Evaluation of Accounting Income Numbers," *Journal of Accounting Research*, 1968

4. William Beaver, "The Information Content of Annual Earnings Announcements," *Journal of Accounting Research*, 1968

5. Rajeev Bhattacharya, *Options and Market Efficiency: A Big Data Approach*, Lambert Academic Publishing, 2020

6. Rajeev Bhattacharya, "Objective Measures of Market Efficiency; Applications to Securities Class Actions and Valuations," *Berkeleley Business Law Journal*, 2019

7. Rajeev Bhattacharya, "Non-Monotonicity of Equilibrium Price," *SSRN*, 2017

8. Rajeev Bhattacharya and Stephen O'Brien, "Arbitrage Risk and Market Efficiency — Applications to Securities Class Actions," *Santa Clara Law Review,* 2015; profiled at Stanford University (http://securities.stanford.edu/resources-academic.html)

9. Lawrence Blume, David Easley and Maureen O'Hara, "Market Statistics and Technical Analysis: The Role of Volume," *Journal of Finance,* 1994

10. Rodney Boehme, Bartley Danielsen and Sorin Sorescu, "Short -Sale Constraints, Differences of Opinion, and Overvaluation," *Journal of Financial and Quantitative Analysis*, 2006

11. Ekkehart Boehmer and Juan (Julie) Wu, "Short Selling and the Price Discovery Process," *Review of Financial Studies*, 2013

12. Ekkehart Boehmer, Zsuzsa Huszar and Bradford Jordan, "The good news in short interest," *Journal of Financial Economics*, 2010

13. Arturo Bris, William Goetzmann and Ning Zhu, "Efficiency and the Bear: Short Sales and Markets Around the World," *Journal of Finance*, 2007

14. Stephen Brown and Jerold Warner, "Using daily stock returns: The case of event studies," *Journal of Financial Economics*, 1985

15. John Campbell, Andrew Lo and A. Craig MacKinlay, *The Econometrics of Financial Markets,* Princeton University Press, Princeton, NJ, 1997

16. Charles Cao, Bing Liang, Andrew Lo and Lubomir Petrasek, "Hedge Fund Holdings and Stock Market Efficiency," *Review of Asset Pricing Studies*, 2018

17. Chin-Han Chiang, Wei Dai, Jianqing Fan, Harrison Hong and Jun Tu, "Robust Measures of Earnings Surprises," *SSRN*, 2016

18. Daniel Collins and S.P. Kothari, "An analysis of intertemporal and cross-sectional determinants of earnings response coefficients," *Journal of Accounting and Economics*, 1989

19. Martijn Cremers and David Weinbaum, "Deviations from Put-Call Parity and Stock Return Predictability," *Journal of Finance and Quantitative Analysis,* 2010

20. Patricia Dechow, Amy Hutton, Lisa Meulbroek and Richard Sloan, "Short-Sellers, Fundamental Analysis, and Stock Returns," *Journal of Financial Economics*, 2001

21. Douglas Diamond and Robert Verrecchia, "Constraints on Short-Selling and Asset Price Adjustment to Private Information," *Journal of Financial Economics*, 1987

22. Itamar Drechsler and Qingyi Drechsler, "The Shorting Premium and Asset Pricing Anomalies," *SSRN*, 2016

23. Ying Duan, Gang Hu and R. David McLean, "Costly arbitrage and idiosyncratic risk: Evidence from short sellers," *Journal of Financial Intermediation,* 2010

24. Darrell Duffie, "Special Repo Rates," *Journal of Finance*, 1996

25. Darrell Duffie, Nicolae Garleanu and Lasse Heje Pedersen, "Securities lending, shorting, and pricing," *Journal of Financial Economics*, 2002

26. David Easley, Maureen O'Hara and P.S. Srinivas, "Option Volume and Stock Prices: Evidence on Where Informed Traders Trade," *Journal of Finance*, 1998

27. Eugene Fama, "Efficient capital markets: II," *Journal of Finance*, 1991

28. Eugene Fama, "Efficient Capital Markets: A Review of Theory and Empirical Work," *Journal of Finance,* 1970

29. Eugene Fama and Kenneth French, "A five-factor asset pricing model," *Journal of Financial Economics*, 2015

30. Eugene Fama and Kenneth French, "Common risk factors in the returns on stocks and bonds," *Journal of Financial Economics*, 1993

31. Eugene Fama and Kenneth French, "The Cross-Section of Expected Stock Returns," *Journal of Finance*, 1992

32. Eugene Fama, Lawrence Fisher, Michael Jensen and Richard Roll, "The Adjustment of Stock Prices to New Information," *International Economic Review*, 1969

33. Michael Goldstein, Edith Hotchkiss and Erik Sirri, "Transparency and Liquidity: A Controlled Experiment on Corporate Bonds," *Review of Financial Studies,* 2007

34. William Greene, *Econometric Analysis,* Pearson, 2018

35. Zvi Griliches and Jerry Hausman, "Errors in Variables in Panel Data," *Journal of Econometrics*, 1986

36. Sanford Grossman and Joseph Stiglitz, "On the Impossibility of Informationally Efficient Markets," *American Economic Review*, 1980

37. Donatien Hainaut and David Colwell, "A structural model for credit risk with switching processes and synchronous jumps," *European Journal of Finance*, 2016

38. Lars Hansen and Kenneth Singleton, "Generalized Instrumental Variables Estimation of Nonlinear Rational Expectations Models," *Econometrica*, 1982

39. Larry Harris, *Trading & Exchanges: Market Microstructure for Practitioners,* Oxford University Press, 2003

40. Jerry Hausman, Gregory Leonard and Douglas Zona, "Competitive Analysis with Differenciated Products," *Annales d'Économie et de Statistique*, 1994

41. Jerry Hausman and William Taylor, "Panel Data and Unobservable Individual Effects," *Econometrica*, 1981

42. Kewei Hou and Tobias Moskowitz, "Market Frictions, Price Delay, and the Cross-Section of Expected Returns," *Review of Financial Studies*, 2005

43. Michael Jensen, "Some Anomalous Evidence Regarding Market Efficiency," *Journal of Financial Economics,* 1978

44. Charles Jones and Owen Lamont, "Short-sale constraints and stock returns," *Journal of Financial Economics*, 2002

45. Jonathan Karpoff, "The Relation Between Price Changes and Trading Volume: A Survey," *Journal of Financial and Quantitative Analysis,* 1987

46. Robert Klemkosky and Bruce Resnick, "Put-Call Parity and Market Efficiency," *Journal of Finance*, 1979

47. S.P. Kothari, "Capital markets research in accounting," *Journal of Economics and Accounting,* 2001

48. S.P. Kothari, Eric So and Rodrigo Verdi, "Analysts' Forecasts and Asset Pricing: A Survey," *Annual Review of Financial Economics*, 2016

49. S.P. Kothari and Jerold Warner, "Econometrics of Event Studies," in Espen Eckbo, ed., *Handbook of Empirical Corporate Finance*, Elsevier/North-Holland, 2007

50. Owen Lamont and Jeremy Stein, "Aggregate Short Interest and Market Valuations," *American Economic Review*, 2004

51. Owen Lamont and Richard Thaler, "Can the Market Add and Subtract? Mispricing in Tech Stock Carve-outs," *Journal of Political Economy*, 2003

52. Baruch Lev, "On the Usefulness of Earnings and Earnings Research: Lessons and Direction from Two Decades of Empirical Research," *Journal of Accounting Research*, 1989

53. Tim Loughran and Jay Ritter, "Uniformly Least Powerful Tests of Market Efficiency," *Journal of Financial Economics,* 2000

54. Burton Malkiel, "Efficient Market Hypothesis," in P. Newman, M. Milgate, and J. Eatwell, eds., *New Palgrave Dictionary of Money and Finance,* Macmillan, London, 1992

55. Burton Malkiel, "The Efficient Market Hypothesis and Its Critics," *CEPS Working Paper,* 2003

56. Harry Markowitz, "The early history of portfolio theory: 1600–1960," *Financial Analysts Journal*, 1999

57. Edward Miller, "Risk, Uncertainty, and Divergence of Opinion," *Journal of Finance*, 1977

58. Merton Miller, "The History of Finance," *The Journal of Portfolio Management*, 1999

59. Randall Morck, Bernard Yeung and Wayne Yu, "The information content of stock markets: why do emerging markets have synchronous stock price movements?" *Journal of Financial Economics*, 2000

60. Dmitriy Muravyev, Neil Pearson and Joshua Pollet, "Is There a Risk Premium in the stock Lending Market?" *SSRN*, 2016

61. Eli Ofek, Matthew Richardson and Robert Whitelaw, "Limited arbitrage and short sales restrictions: evidence from the options markets," *Journal of Financial Economics*, 2004

62. Maureen O'Hara, *Market Microstructure Theory,* Blackwell Publishing, Malden, MA, 1997

63. Jeffrey Pontiff, "Costly arbitrage and the myth of idiosyncratic risk," *Journal of Accounting and Economics,* 2006

64. Rafael Porta, Josef Lakonishok, Andrei Shleifer and Robert Vishny, "Good News for Value Stocks: Further Evidence on Market Efficiency," *NBER Working Paper,* 1995

65. David Rapach, Matthew Ringgenberg and Guofu Zhou, "Short interest and aggregate stock returns," *Journal of Financial Economics*, 2016

66. Michael Roberts and Toni Whited, "Endogeneity in Empirical Corporate Finance,". in M. Constantinides and R. Stulz, eds., *Handbook of the Economics of Finance*, 2011

67. Richard Roll, "A Simple Implicit Measure of the Effective Bid-Ask Spread in an Efficient Market," *The Journal of Finance,* 1984

68. Pedro Saffi and Kari Sigurdsson, "Price Efficiency and Short Selling," *Review of Financial Studies*, 2011

69. Paul Samuelson, "An Enjoyable Life Puzzling Over Modern Finance Theory," *Annual Review of Financial Economics,* 2009

70. Hersh Shefrin, *A Behavioral Approach to Asset Pricing*, Academic Press, 2008

71. Hersh Shefrin, "Building On Kahneman's Insights in the Development of Behavioral Finance," *Loyola University Chicago Law Journal*, 2013

72. William Sharpe, "Capital asset prices: A theory of market equilibrium under conditions of risk," *Journal of Finance*, 1964

73. Andrei Shleifer, *Inefficient Markets — An Introduction to Behavioral Finance*, Oxford University Press, 2000

74. Robert Shiller, "The Use of Volatility Measures in Assessing Market Efficiency," *NBER Working Paper,* 1981

75. Scott Stickel and Robert Verrecchia, "Evidence that Trading Volume Sustains Stock Price Changes," *Financial Analysts Journal,* 1994

76. Jiang Wang, "A Model of Competitive Stock Trading Volume," *Journal of Political Economy,* 1994

77. Babajide Wintoki, James Linck and Jeffrey Netter, "Endogeneity and the Dynamics of Internal Corporate Governance," *Journal of Financial Economics*, 2012

78. Jeffrey Wooldridge, *Econometric Analysis of Cross Section and Panel Data*, MIT Press, 2010

79. Jeffrey Wurgler and Ekaterina Zhuravskaya, "Does Arbitrage Flatten Demand Curves for Stocks," *Journal of Business*, 2002

Appendix 1

Summary Statistics of Negative Measures of Market Efficiency

All Stocks

Statistic	Variable	KDAR (CAPM)	KDAR (MM)	EAAR (CAPM)	EAAR (MM)	Arb Risk (CAPM)	Arb Risk (MM)	Idio Risk (CAPM)	Idio Risk (MM)
MEAN		0.01625	0.01558	0.03272	0.03083	0.02967	0.02917	0.04190	0.04111
STD		0.02749	0.02689	0.03956	0.03712	0.02583	0.02564	0.03877	0.03866
N		62,169	62,169	88,450	81,196	99,048	93,034	101,500	101,500
CORR	**KDAR (CAPM)**	1.00000	0.96014	0.36935	0.32078	0.40029	0.39016	0.37721	0.38367
CORR	**KDAR (MM)**	0.96014	1.00000	0.37495	0.35727	0.44469	0.43608	0.40969	0.41894
CORR	**EAAR (CAPM)**	0.36935	0.37495	1.00000	0.87660	0.43598	0.43061	0.41597	0.41960
CORR	**EAAR (MM)**	0.32078	0.35727	0.87660	1.00000	0.48273	0.47657	0.45980	0.46869
CORR	**Arb Risk (CAPM)**	0.40029	0.44469	0.43598	0.48273	1.00000	0.99638	0.79794	0.81006
CORR	**Arb Risk (MM)**	0.39016	0.43608	0.43061	0.47657	0.99638	1.00000	0.79975	0.81221
CORR	**Idio Risk (CAPM)**	0.37721	0.40969	0.41597	0.45980	0.79794	0.79975	1.00000	0.99288
CORR	**Idio Risk (MM)**	0.38367	0.41894	0.41960	0.46869	0.81006	0.81221	0.99288	1.00000

Appendix 1

Summary Statistics of Negative Measures of Market Efficiency

All Nasdaq Stocks

Statistic	Variable	KDAR (CAPM)	KDAR (MM)	EAAR (CAPM)	EAAR (MM)	Arb Risk (CAPM)	Arb Risk (MM)	Idio Risk (CAPM)	Idio Risk (MM)
MEAN		0.01947	0.01901	0.03764	0.03658	0.03490	0.03430	0.04793	0.04755
STD		0.03236	0.03180	0.04459	0.04144	0.02784	0.02772	0.04160	0.04160
N		38,740	38,740	54,281	48,665	62,068	58,030	63,854	63,854
CORR	**KDAR (CAPM)**	1.00000	0.97003	0.35788	0.30717	0.37058	0.35949	0.36452	0.36673
CORR	**KDAR (MM)**	0.97003	1.00000	0.36004	0.32954	0.40877	0.39910	0.39071	0.39391
CORR	**EAAR (CAPM)**	0.35788	0.36004	1.00000	0.87490	0.41088	0.40491	0.40131	0.40186
CORR	**EAAR (MM)**	0.30717	0.32954	0.87490	1.00000	0.44754	0.44040	0.43151	0.43394
CORR	**Arb Risk (CAPM)**	0.37058	0.40877	0.41088	0.44754	1.00000	0.99594	0.80462	0.80988
CORR	**Arb Risk (MM)**	0.35949	0.39910	0.40491	0.44040	0.99594	1.00000	0.80654	0.81209
CORR	**Idio Risk (CAPM)**	0.36452	0.39071	0.40131	0.43151	0.80462	0.80654	1.00000	0.99683
CORR	**Idio Risk (MM)**	0.36673	0.39391	0.40186	0.43394	0.80988	0.81209	0.99683	1.00000

Appendix 1

Summary Statistics of Negative Measures of Market Efficiency

All Non-Nasdaq Stocks

Statistic	Variable	KDAR (CAPM)	KDAR (MM)	EAAR (CAPM)	EAAR (MM)	Arb Risk (CAPM)	Arb Risk (MM)	Idio Risk (CAPM)	Idio Risk (MM)
MEAN		0.01093	0.00990	0.02489	0.02222	0.02089	0.02065	0.03167	0.03019
STD		0.01510	0.01395	0.02814	0.02733	0.01905	0.01893	0.03083	0.03009
N		23,429	23,429	34,169	32,531	36,980	35,004	37,646	37,646
CORR	**KDAR (CAPM)**	1.00000	0.87364	0.33186	0.25994	0.42850	0.43089	0.35195	0.36714
CORR	**KDAR (MM)**	0.87364	1.00000	0.34232	0.34961	0.51666	0.52233	0.42198	0.45109
CORR	**EAAR (CAPM)**	0.33186	0.34232	1.00000	0.86520	0.42317	0.42850	0.38923	0.39597
CORR	**EAAR (MM)**	0.25994	0.34961	0.86520	1.00000	0.47826	0.48235	0.45940	0.48100
CORR	**Arb Risk (CAPM)**	0.42850	0.51666	0.42317	0.47826	1.00000	0.99678	0.73426	0.76476
CORR	**Arb Risk (MM)**	0.43089	0.52233	0.42850	0.48235	0.99678	1.00000	0.73710	0.76832
CORR	**Idio Risk (CAPM)**	0.35195	0.42198	0.38923	0.45940	0.73426	0.73710	1.00000	0.97969
CORR	**Idio Risk (MM)**	0.36714	0.45109	0.39597	0.48100	0.76476	0.76832	0.97969	1.00000

Appendix 2

Summary Statistics of All Potential Factors

All Nasdaq Stocks

Statistic	Variable	Norm Volume	Log Market Cap	Rel Bid-Ask Spread	Norm Short Interest	Number Analysts	Dispersion Analysts	Num Market Makers
MEAN		0.00775	18.77403	0.02391	0.04117	4.6	247.9	24.9
STD		0.01095	1.75222	0.03066	0.05409	5.0	14,932.7	15.9
N		63,854	63,854	63,854	28,472	48,234	37,385	63,854
CORR	**Norm Volume**	1.00000	0.21829	-0.19259	0.43856	0.32661	-0.00235	0.34081
CORR	**Log Market Cap**	0.21829	1.00000	-0.68426	0.35660	0.71989	-0.01716	0.69267
CORR	**Rel Bid-Ask Spread**	-0.19259	-0.68426	1.00000	-0.32742	-0.38571	0.01894	-0.54042
CORR	**Norm Short Interest**	0.43856	0.35660	-0.32742	1.00000	0.27012	-0.00778	0.46306
CORR	**Number Analysts**	0.32661	0.71989	-0.38571	0.27012	1.00000	-0.00998	0.71647
CORR	**Dispersion Analysts**	-0.00235	-0.01716	0.01894	-0.00778	-0.00998	1.00000	-0.01226
CORR	**Num Market Makers**	0.34081	0.69267	-0.54042	0.46306	0.71647	-0.01226	1.00000

Appendix 2

Summary Statistics of All Potential Factors

All Non-Nasdaq Stocks

Statistic	Variable	Norm Volume	Log Market Cap	Rel Bid-Ask Spread	Norm Short Interest	Number Analysts	Dispersion Analysts
MEAN		0.00693	20.51421	0.01442	0.03964	7.8	0.4
STD		0.00831	2.11559	0.02420	0.07451	6.5	19.2
N		37,646	37,646	37,646	24,394	32,456	28,987
CORR	**Norm Volume**	1.00000	0.18301	-0.23188	0.32338	0.27468	0.00272
CORR	**Log Market Cap**	0.18301	1.00000	-0.57445	-0.04683	0.72657	-0.01154
CORR	**Rel Bid-Ask Spread**	-0.23188	-0.57445	1.00000	-0.04627	-0.33291	0.02033
CORR	**Norm Short Interest**	0.32338	-0.04683	-0.04627	1.00000	0.02011	0.07101
CORR	**Number Analysts**	0.27468	0.72657	-0.33291	0.02011	1.00000	-0.01296
CORR	**Dispersion Analysts**	0.00272	-0.01154	0.02033	0.07101	-0.01296	1.00000